SIGN-LANGUAGE

Spring 2026

Meeting No. 6

System Development

& Datasets

SYSTEM UPDATES

Compare the performance of SlowFast vs. ViViT on a custom 10-class sign language dataset.

Both models were pre-trained on the massive Kinetics-400 dataset and have conceptually the same weights

To ensure that it would be “fair” to compare them , video with signs was prerecorded

Имя	Дата изменения	Тип	Размер
a	31.03.2026 18:01	Папка с файлами	
a lot	31.03.2026 18:01	Папка с файлами	
abdomen	31.03.2026 18:01	Папка с файлами	
able	31.03.2026 18:01	Папка с файлами	
about	31.03.2026 18:01	Папка с файлами	
above	31.03.2026 18:01	Папка с файлами	
accent	31.03.2026 18:01	Папка с файлами	
accept	31.03.2026 18:01	Папка с файлами	
accident	31.03.2026 18:01	Папка с файлами	
accomplish	31.03.2026 18:01	Папка с файлами	

SYSTEM UPDATES (SLOW-FAST)

Training Accuracy climbed steadily: 16.3% $\rightarrow$ 73.5%.

Validation Accuracy peaked at 50.0% at Epoch 9.

Epoch 05 | train_loss=1.2914 train_acc=61.2% | val_loss=1.9590 val_acc=40.0%
Saved model to: D:\Projects_Python\Senior\video\slowfast_finetuned
Epoch 06 | train_loss=1.4025 train_acc=46.9% | val_loss=2.3810 val_acc=20.0%
Epoch 07 | train_loss=1.2561 train_acc=57.1% | val_loss=2.1247 val_acc=40.0%
Saved model to: D:\Projects_Python\Senior\video\slowfast_finetuned
Epoch 08 | train_loss=1.0676 train_acc=65.3% | val_loss=2.1865 val_acc=40.0%
Saved model to: D:\Projects_Python\Senior\video\slowfast_finetuned
Epoch 09 | train_loss=0.9199 train_acc=73.5% | val_loss=2.0877 val_acc=50.0%

custom testing videos per class were created (examples to the right)

SYSTEM UPDATES (SLOW-FAST)

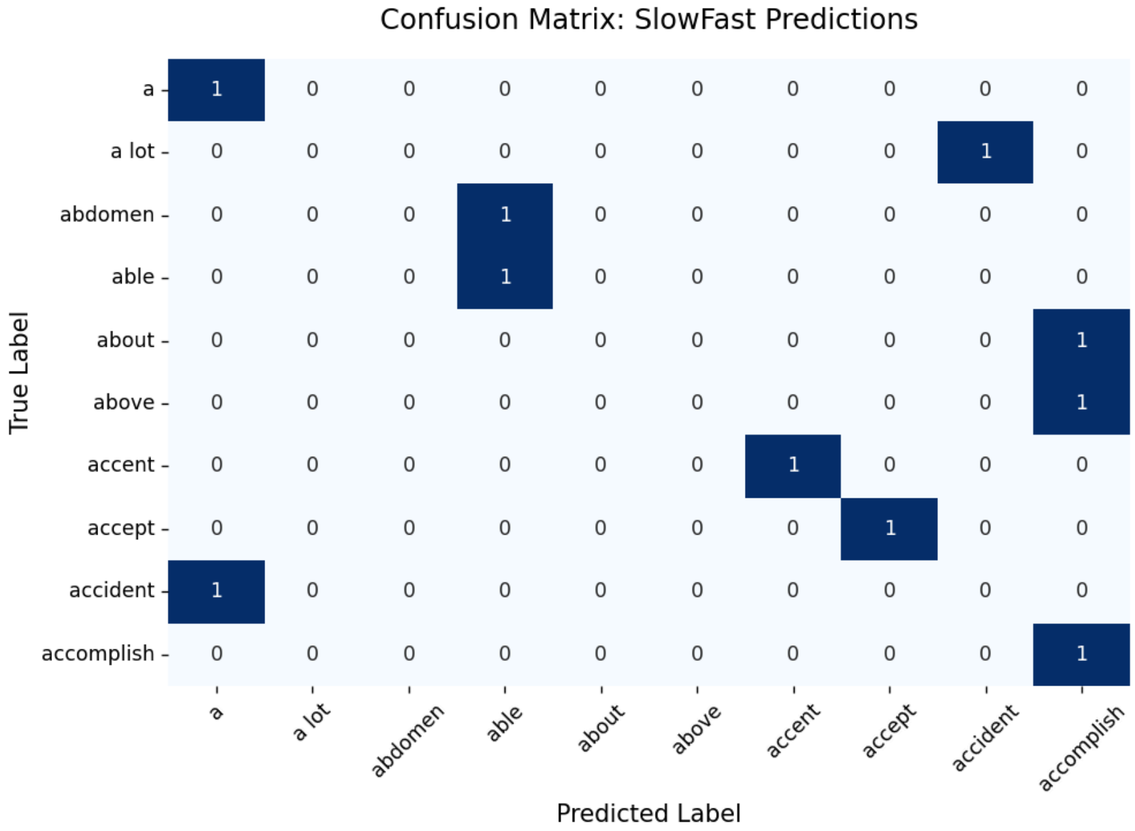
Fast Pathway: torch.arange(0, 64, 2)

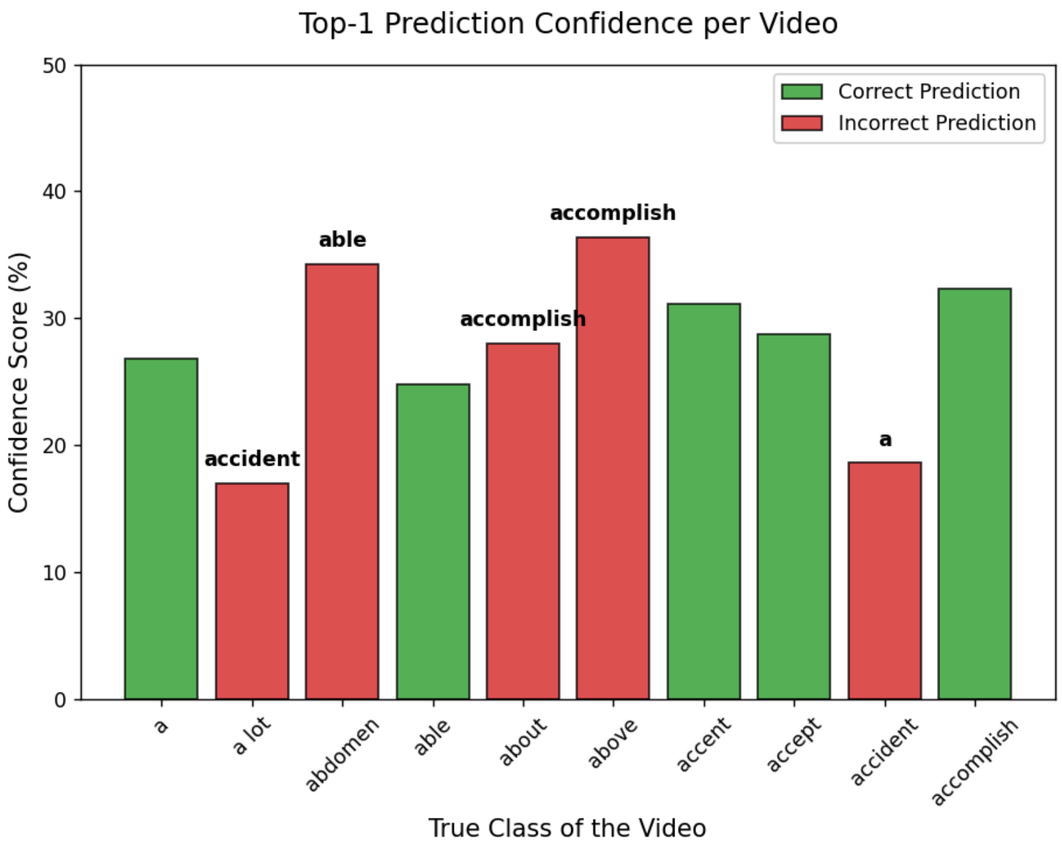
- It takes every 2nd frame from your 64-frame subset.
- Result: 32 frames total (64 / 2 = 32).
- Purpose: Captures motion data (low resolution usually, but here high temporal frequency).

Slow Pathway: torch.arange(0, 64, 8)

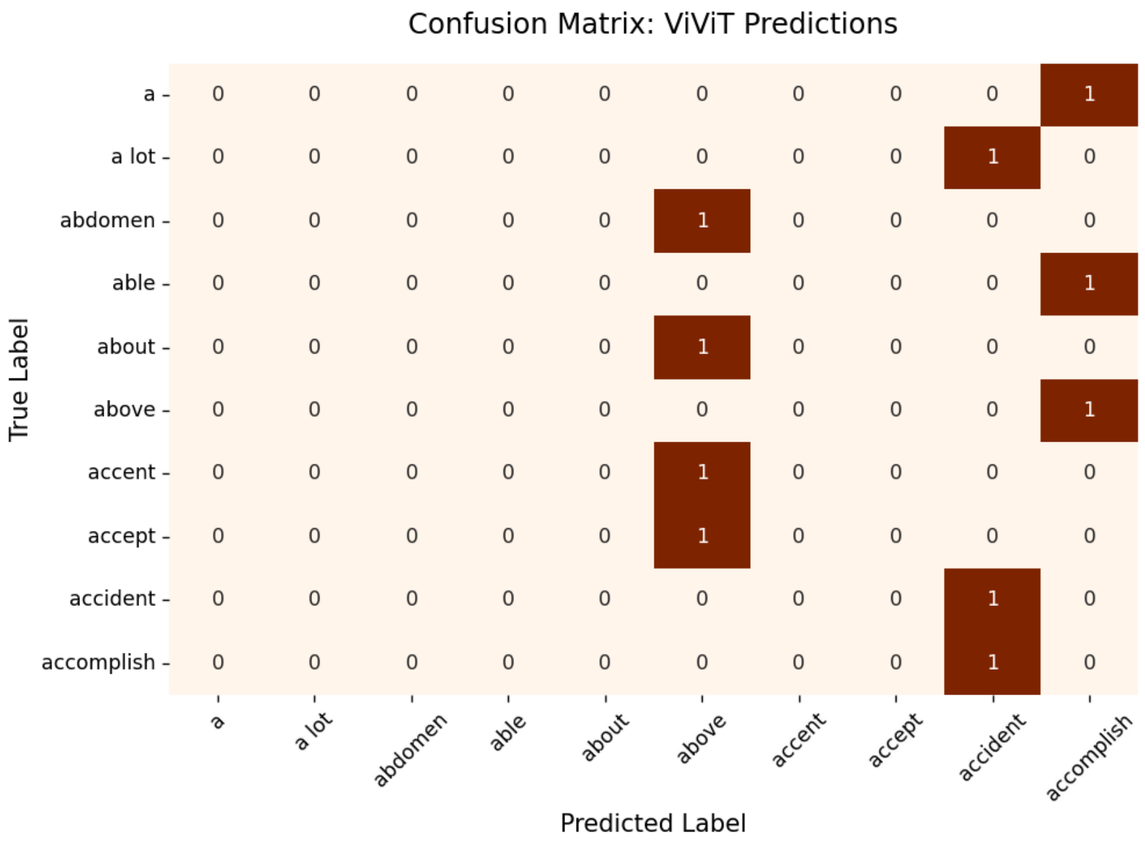
- It takes every 8th frame from your 64-frame subset.
- Result: 8 frames total (64 / 8 = 8).
- Purpose: Captures spatial data (high resolution, low temporal frequency).

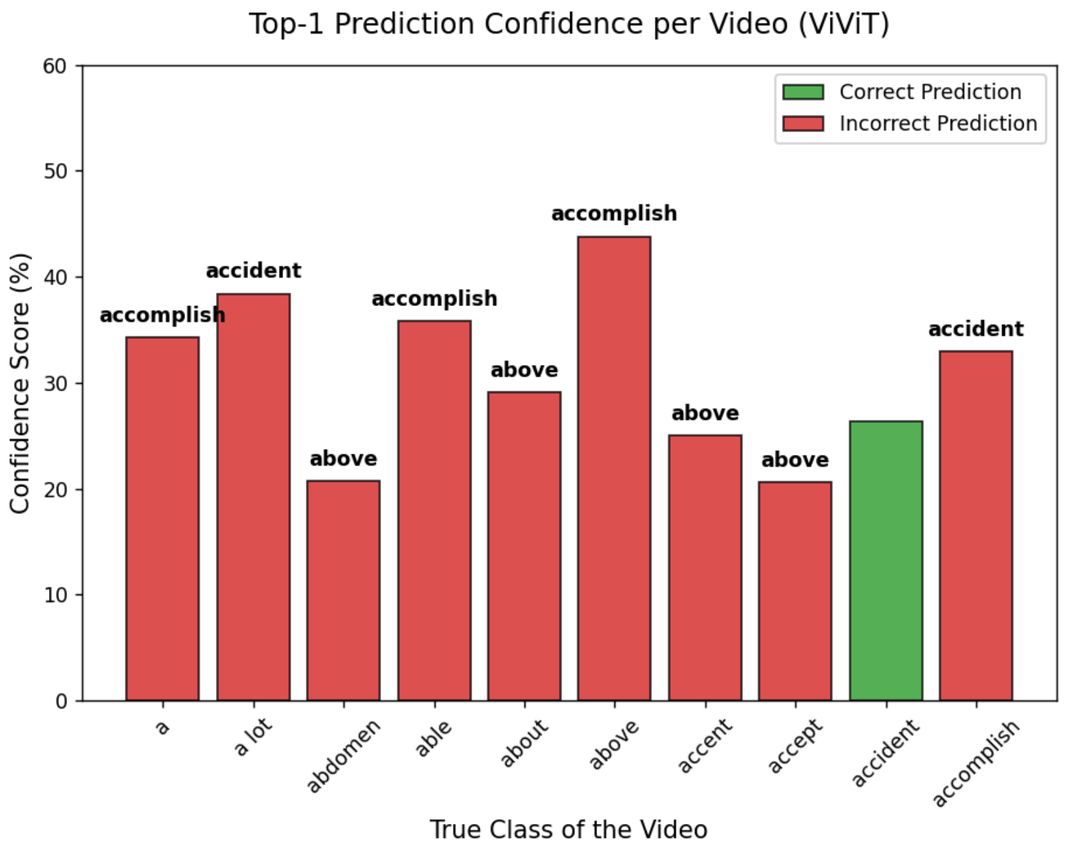
SLOWFAST RESULTS



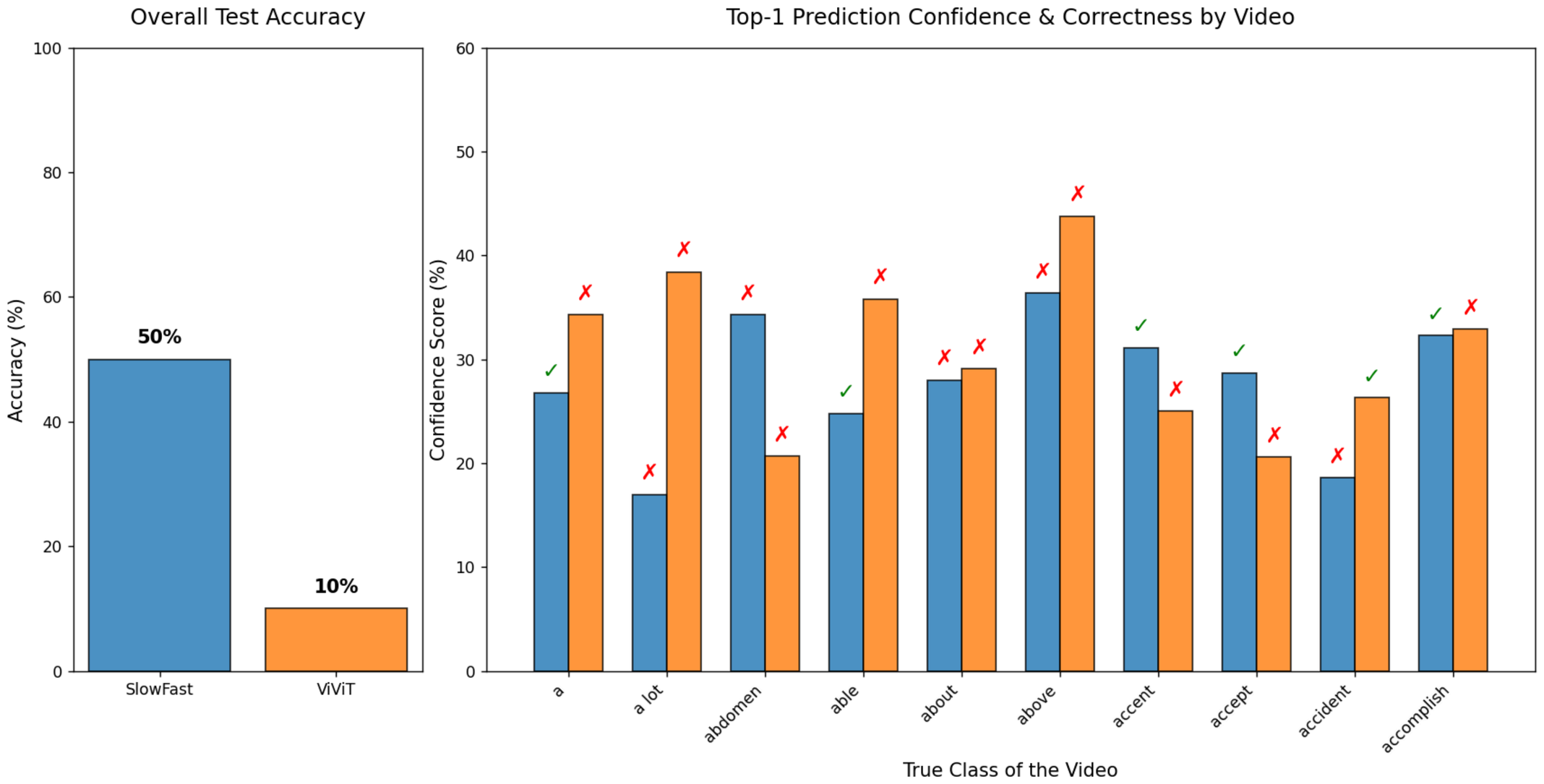


VIVIT RESULTS





COMPARISON


Overall Test Accuracy
Accuracy (%)
100
80
60
40
20
0
50%
10%
SlowFast
ViViT
Top-1 Prediction Confidence & Correctness by Video
Confidence Score (%)
60
50
40
30
20
10
0
a
a lot
abdomen
able
about
above
accent
accept
accident
accomplish
True Class of the Video

ViViT Frame Sampling in the Paper

Training: Uniformly sample 32 frames from the input video clip.
Inference: 32 frames using a stride of 2, get total of 4 inputs per video.

```
def _sample_frames(self, frames):
    idx = np.linspace(0, len(frames) - 1, self.num_frames).astype(int)
    return [frames[i] for i in idx]
```

ViViT Fine-Tuning Results With New Frame Sampling Method

ViViT	WLASL-100	WLASL-300
Full Fine-Tuning	55.81%	36.98%
LoRA Fine-Tuning	53.10%	7.19%

Slow Pathway in SlowFast

Slow pathway purpose: capture semantic information, high-level spatial features that don't change quickly.
Temporal reduction: Instead of processing all frames, we sample fewer frames .
Typical ratio: In SlowFast, alpha = 4, meaning Slow pathway is 1/4 the frame rate of Fast pathway.

We have 64 frames clip input

Fast pathway: 32 frames (captures motion)
Slow pathway: 32 / 4 = 8 frames (captures context)

Why to choose such stride?
8 frames are enough to capture overall action context without losing important information in sign language recognition.
We use configuration of slowfast model 8x8 (Frames for Slow Pathway x temporal stride)

FuseFastToSlow Module:

- Fast features are temporally downsampled using a 3D convolution with stride = alpha.
- Convolution reduces temporal length of Fast features from 32 → 8 frames to match Slow pathway.
- Resulting Fast features are projected to the same channel dimension as Slow features

GitHub Link:

AbdollaZ/
seniorproject_signlangu...
This is a repository for hosting CSCI 409 Senior Project documentation and files. Project topic is Sign Language Capturing via...
4
Contributors
0
Issues
0
Stars
0
Forks
AbdollaZ/seniorproject_signlanguage: This is a repository for hosting CSCI 409 Senior Project documentation and files. Project topic is...

THANK YOU